Name	Type	Feature of Interest	Property	Characteristics	Components	Justification
The name of each IoT device	E.g., sensor, actuator, or device	What "thing" is being observed and/or acted upon?	Which aspect(s) of the "thing" is/are being observed and/or acted upon?	What are the characteristics of the device?	What are the components embedded in the device?	Why is the device selected for the proposed IoT solution?
DFRobot DS18B20 Waterproof Temperature Sensor (BOS0042) / 防水温度传感器	Sensor	Water in each breeding tank	Water temperature	Waterproof DS18B20 probe; 3–5 V; PH2.0-3P connector; sensing range −55 to 125 °C; display range −10 to 85 °C with ±0.5 °C stated error; requires a pull-up resistor. Source: DFRobot BOS0042.	Waterproof DS18B20 probe, 3-pin cable, pull-up resistor, PH2.0 adapter and controller input.	Direct temperature feedback supports heater control and alarm limits in wet tanks. The sealed probe is suitable for remote or humid measurement, while local validation prevents unsafe heating commands.
DFRobot SEN0189 Turbidity Sensor / 浑浊度传感器	Sensor	Water clarity in each breeding tank	Turbidity	Optical infrared emitter/receiver; 5 VDC; 40 mA maximum; response below 500 ms; selectable analogue 0–4.5 V or thresholded digital output; operating range −30 to 80 °C. The probe top is not waterproof.	Optical turbidity probe, interface board, A/D or GPIO input, analogue/digital selector and threshold potentiometer.	Provides an early warning of suspended solids, filtration faults and contamination. Digital threshold alarms can be local, while analogue values support trend analysis and water-change decisions.
DFRobot SEN0161-V2 Gravity Analog pH Meter / 模拟 pH 计 V2	Sensor	Water chemistry in each breeding tank	pH value	Signal board supply 3.3–5.5 V; analogue output 0–3.0 V; BNC electrode and PH2.0-3P signal connector; stated accuracy ±0.1 at 25 °C; electrode range pH 0–14, 5–60 °C, response below 2 min; 1 m cable.	Laboratory-grade pH electrode, V2 signal-conversion board, BNC cable, PH2.0-3P lead, calibration buffers and analogue input.	pH is a key water-quality variable for aquaculture. Two-point calibration and local plausibility/rate checks allow the PLC to alarm and bound corrective actions.
Lonhand LH-WQS-AN-485-UI NH3-N Ion Sensor / 氨氮离子传感器	Sensor	Water chemistry in each breeding tank	Ammonia-nitrogen concentration	Measurement range 0.1–18,000 mg/L; temperature range 0–60 °C; 4–20 mA and RS-485 outputs; IP68; 5 m cable; NPT3/4 or M39×1.5 connection; 0–3 bar; pH 4–10; repeatability ± 4%; PC/PBT corrosion-resistant housing.	Ion-selective ammonia electrode, 4–20 mA/RS-485 transmitter, 5 m cable, immersion fitting, PLC analogue/Modbus input and calibration solution.	Ammonia is a high-risk water-quality condition. A wired industrial output supports continuous trend logging, local alarm thresholds and dependable response even when the network is unavailable.
AXIS M3086-V Network Camera / 网络摄像机	Sensor	Breeding and nesting-zone activity and evidence	Video stream and event evidence	4 MP maximum resolution 2688×1512; 2.4 mm F2.1 lens; 130° horizontal and 93° vertical field of view; H.264/H.265/Motion JPEG; 25/30 fps; WDR up to 120 dB; Lightfinder; PoE Class 2; IP42 and IK08 variants.	Network camera, lens, PoE RJ45 connector, SD-card slot, control button, mounting dome and managed-switch port.	Video provides farm-wide context and event evidence for activity, feeding and nesting. PoE simplifies installation; edge analytics should retain only validated clips rather than upload continuous video.
Raspberry Pi 5 (8 GB)	Device	Edge gateway and local sensor aggregation	Local processing, protocol bridging and storage	Broadcom BCM2712 2.4 GHz quad-core 64-bit Arm Cortex-A76; 8 GB LPDDR4X-4267; standard 40-pin GPIO header; USB, Ethernet, Wi-Fi and Bluetooth; 5 V USB-C power and active cooling recommended for sustained workloads.	Raspberry Pi 5 board, microSD or SSD, 40-pin GPIO interface, USB/Ethernet/Wi-Fi links, power supply, case and active cooler.	A low-cost Linux edge node can bridge sensors, MQTT and dashboards, buffer data and pre-process camera events. Safety-critical interlocks remain in the dedicated PLC rather than the general-purpose board.
ESP32-WROOM-32	Device	Distributed sensor and low-power actuator nodes	Wireless telemetry and GPIO control	Dual-core Xtensa LX6 up to 240 MHz; 448 KB ROM, 520 KB SRAM and 4 MB SPI flash; 2.4 GHz 802.11 b/g/n up to 150 Mbps; Bluetooth 4.2/BLE; up to 32 GPIO; ADC, DAC, touch, PWM and serial peripherals; 3.0–3.6 V supply.	ESP32-WROOM-32 module, PCB antenna, 40 MHz crystal, flash, GPIO/ADC interface, regulated 3.3 V supply and enclosure.	Provides an economical wireless interface for non-critical distributed measurements. Wet-area deployments still need isolation, watchdogs and bounded local outputs before any actuator command is accepted.
Ubuntu Server 22.04	Device	Edge/server software platform	Container, storage, MQTT and API runtime	Ubuntu Server 22.04 LTS provides a maintainable Linux base for scale-out services, networking, storage, containers and system administration. The workload can run on Raspberry Pi 5 or another supported ARM/x86 host.	Ubuntu Server image, host storage, network interface, container runtime, MQTT broker, time-series database, API service and monitoring.	A documented LTS platform supports repeatable deployment, security updates and service separation for the edge gateway. It improves maintainability without placing deterministic safety control in the operating system.
HC-SR501 PIR Motion Sensor / 人体红外传感器	Sensor	Activity in breeding or nesting areas	Motion event	Pyroelectric dual-element PIR module; 4.5–20 V supply; below 50 µA stated current; 3.3 V high/0 V low GPIO output; cone below 100°; three pins; selectable repeatable/non-repeatable trigger and adjustable delay/sensitivity.	PIR element, Fresnel lens, comparator board, delay and sensitivity potentiometers, VCC/OUT/GND header and GPIO input.	A low-cost motion trigger can wake or tag camera recording and reduce continuous processing. It is not a safety sensor because initialization time, wind, lighting and animal movement can cause false or missed events.
PondMAX Solar Automatic Fish Feeder	Actuator	Feed hopper and pond feeding schedule	Scheduled feed dispensing	Solar charging with built-in 2200 mAh lithium battery; 1–6 feedings per day; adjustable S/M/L throw and 0–20 s feed duration; 3–10 mm food size; 12 V/1.66 A adapter specification in the supplied manual.	Solar panel, rechargeable battery, hopper, motor and spreader, timer/display controls, protective cover and mounting bracket.	Automated feeding improves schedule consistency and reduces manual labour. Feed commands should remain bounded by time, quantity and hopper checks, with an operator override and an audit record.
Bluefish MINI-200 Submersible Pump / 潜水泵	Actuator	Tank recirculation and water-change line	Water circulation and flow	12 VDC; 0.4 A; 4.8 W; 200 L/h open flow; 1.5 m maximum head; brushless motor; IP68 submersible construction; approximately 43×38×30 mm; below 35 dB; dry-run/block protection options.	Brushless pump, impeller, inlet/outlet, sealed motor, silicone vibration feet, 12 V supply, relay/MOSFET driver and flow feedback.	Low-voltage circulation supports filtration and controlled water changes. A relay, flow confirmation and dry-run protection prevent a blocked line or empty tank from damaging the pump.